

Harsh Raj

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SUMMARY

Cloud-native engineer working where Kubernetes internals meet security: controllers, admission and policy engines, workload identity and supply chain. 14 pull requests across 7 CNCF projects, 3 merged in KEDA and Kyverno, and 15,000+ lines shipped across an admission webhook, a multimodal detection model and a protocol scanner. Open to internships and contract work.

EDUCATION

UPES Dehradun

2023 - 2027

Bachelor of Technology - Computer Science and Engineering (Cybersecurity and Forensics)

TECHNICAL SKILLS

- **Languages:** Go, Python, C++, Java, SQL, Bash, JavaScript
- **Cloud Native:** Kubernetes, controller-runtime, Admission Webhooks, CRDs, Docker, containerd, Helm, kind, k3s, Envoy, gRPC / Protobuf, Prometheus, Argo Workflows, Linux, GitHub Actions, CI/CD
- **Security and Supply Chain:** OAuth 2.0, OIDC, JWT, Workload Identity, PKI, Sigstore, Cosign, TUF, Nmap, OpenVAS, Burp Suite, Wireshark, OWASP Top 10, SIEM, Incident Response, Digital Forensics
- **Machine Learning:** PyTorch, Hugging Face Transformers, scikit-learn, CNNs, Siamese Networks, Multimodal Fusion, Adversarial Robustness, SHAP, Grad-CAM, LLMs, RAG
- **Testing and Web:** Ginkgo / Gomega, pytest, Playwright, PostgreSQL, Redis, SQLite, FastAPI, REST APIs, React

EXPERIENCE

Software Developer Intern - IBM

Jun 2026 - Jul 2026

- Built a multilingual disaster-response assistant on a RAG pipeline over LLaMA 3.2 with speech-to-text and text-to-speech, plus a one-click risk report pulling geolocation, air quality, weather and disaster data.

Stack: Python, Streamlit, LLaMA 3.2, RAG, Gemini, REST APIs

Cybersecurity Intern - Safepay

Jun 2025 - Jul 2025

- Triageed 50+ SIEM alerts by correlating event metadata, affected assets, severity and indicators of compromise.
- Ran Nmap and OpenVAS vulnerability assessments that surfaced 15+ exposed services and misconfigurations, and wrote the triage playbooks the security team adopted.

Stack: SIEM, Nmap, OpenVAS, Wireshark, Linux

OPEN SOURCE CONTRIBUTIONS

Fixes in CNCF projects (github.com/harshrajdebug)

2026

- **KEDA (merged #8193):** fixed a connection leak in the external scaler's gRPC pool, where every scaler address ever configured held a connection and a goroutine for the life of the process; measured on a real API server (held connections 5, 10, 15 across three cycles, down to 0). Follow-up test hardening in #8213.
- **Kyverno (merged #17608, backported #17612):** the CLI resolved no resource kinds for namespaced validating policies, so they ran against an empty set and reported a clean pass while checking nothing. Open fix (#17620) for a panic where one policy's JSON patch crashed the mutating webhook.
- **PipeCD:** restarting piped mid-deployment cancelled the deployment and rolled it back (#7443); a cancelled Terraform stage left the remote state lock held (#7440); ECS rollbacks reported synced while running the old revision (#7412).
- **kgateway, KubeEdge, OpenKruise, Volcano:** TLS policies reported Accepted while the gateway rejected the backend (#14749, #14760); offline edge nodes unable to start pods (#7306, #7307); claim labels overwriting controller labels (#1001); HyperNode matching (#6002).

Stack: Go, Kubernetes, controller-runtime, Envoy, gRPC, TLS, AWS ECS, Terraform

PROJECTS

sigstore-guard - Kubernetes Image Signature Admission Webhook

2026

- Validating admission webhook (~4,000 lines of Go, Apache-2.0, green CI) that rejects pods whose images carry no Sigstore signature from an accepted signer, with the trust root anchored in TUF so key rotation reaches clusters without a redeploy.
- Separated the API server's failure policy from the policy's own fail-open option, the two controls that decide whether the webhook protects a cluster or takes it down.

Stack: Go, Kubernetes, Sigstore, Cosign, TUF, kind

Darkscan - Tor Hidden Service Scanner and Forensics Toolkit

2026

- 2,700-line concurrent scanner that fingerprints .onion services across seven protocols (HTTP, SSH, TLS, FTP, SMTP, IRC, VNC) through a Tor SOCKS5 proxy, probing sensitive paths against a randomized soft-404 baseline, with a dependency-free EXIF parser that recovers GPS and camera metadata.

Stack: Go, SQLite, Tor SOCKS5, goquery, REST API

PhishArmor - Multimodal Deep Learning Phishing Detection

2026

- 8,600-line detector fusing a character-level URL CNN, a DistilBERT text encoder and a ResNet-18 Siamese visual encoder, with SHAP and Grad-CAM explanations behind a FastAPI service and a Chrome extension.
- Caught a 37.3% duplicate-crop leak in the Phishpedia split, correcting top-1 retrieval accuracy from 0.859 to 0.668.

Stack: Python, PyTorch, Hugging Face Transformers, scikit-learn, FastAPI